

CPSC 250

Simple Linear Regression with StatsModels

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1 Motivation

Previously, we used `scipy.optimize.curve_fit()` for fitting data.

In this lecture, we introduce a different approach using:

- `pandas`
- `statsmodels`

The goals are:

1. obtain parameter uncertainties automatically,
2. use a cleaner data structure,
3. simplify regression workflows.

2 Why Pandas?

A `pandas DataFrame` is similar to a spreadsheet.

Each column has a label and stores related data.

This is often easier to work with than raw NumPy matrices.

3 Creating a DataFrame

We first import `pandas`:

```
import pandas as pd
```

Then create a `DataFrame`:

```
df = pd.DataFrame({  
    "x": [5, 10, 15, 20],  
    "y": [5, 12, 20, 27]  
})
```

The keys:

```
"x"  
"y"
```

become the column labels.

4 Accessing Columns

We can print the entire table:

```
print(df)
```

Or access a specific column:

```
print(df["x"])
```

This returns only the x-values.

5 Preparing for Linear Regression

Suppose we want a linear model:

$$y = \beta_0 + \beta_1 x$$

where:

- β_0 is the intercept,
- β_1 is the slope.

6 Importing StatsModels

We typically import:

```
import statsmodels.api as sm
```

7 Extracting the Data

We extract columns from the DataFrame:

```
y = df["y"]  
X = df["x"]
```

Notice:

- uppercase X is used for independent variables,
- lowercase y is used for dependent variables.

8 Adding a Constant Term

We now call:

```
X = sm.add_constant(X)
```

This adds a column of ones.

Why?

Because the model:

$$y = \beta_0 + \beta_1 x$$

contains a constant term.

Without this step, the fit would assume:

$$y = \beta_1 x$$

which forces the line through the origin.

9 Creating the Model

We construct the model using Ordinary Least Squares (OLS):

```
model = sm.OLS(y, X).fit()
```

Important:

- the dependent variable y comes first,
- the independent-variable matrix X comes second.

10 What is OLS?

OLS stands for:

Ordinary Least Squares

This is the standard linear-regression technique.

The idea is to choose parameters that minimize:

$$\sum_i (y_i - y_{\text{fit},i})^2$$

the sum of squared residuals.

11 Viewing the Results

The most useful step is:

```
print(model.summary())
```

This produces a detailed statistical summary including:

- best-fit parameters,
- uncertainties,
- confidence intervals,
- goodness-of-fit measures,
- statistical tests.

12 Advantages of StatsModels

Compared with simpler fitting approaches, `statsmodels` provides:

- cleaner statistical output,
- automatic uncertainty calculations,
- convenient regression tools,
- integration with pandas.

13 Complete Example

```
import pandas as pd
import statsmodels.api as sm

df = pd.DataFrame({
    "x": [5, 10, 15, 20],
    "y": [5, 12, 20, 27]
})

y = df["y"]
X = df["x"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

14 Key Takeaways

- Pandas DataFrames organize data using labeled columns.
- StatsModels provides advanced regression tools.
- `sm.add_constant()` adds the intercept term.
- OLS performs standard least-squares regression.
- `model.summary()` gives detailed statistical information.